

SUPPLY CHAIN ANALYTICS PROJECT

Inventory Misalignment & Network Intelligence System

A multi-dimensional supply chain intelligence framework
for detecting stockout risk, phantom inventory, demand
decay,
supply chain nervousness, and network-level transshipment
optimisation across a 897-store retail network.

4.5M

Rows Processed

Apr–Jun 2024 · 19 cols

\$112,931

Gross Rev. at Risk

Before substitution

\$65,116

Net Rev. at Risk

After 42.3% absorbed

897

Stores Analysed

Multi-city network

Dataset 4,500,000 rows × 19 columns (Apache Parquet)

Period April – June 2024 · Multi-city Retail Network

Stack Python · Pandas · NumPy · Scikit-learn · Power BI

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1 Executive Summary

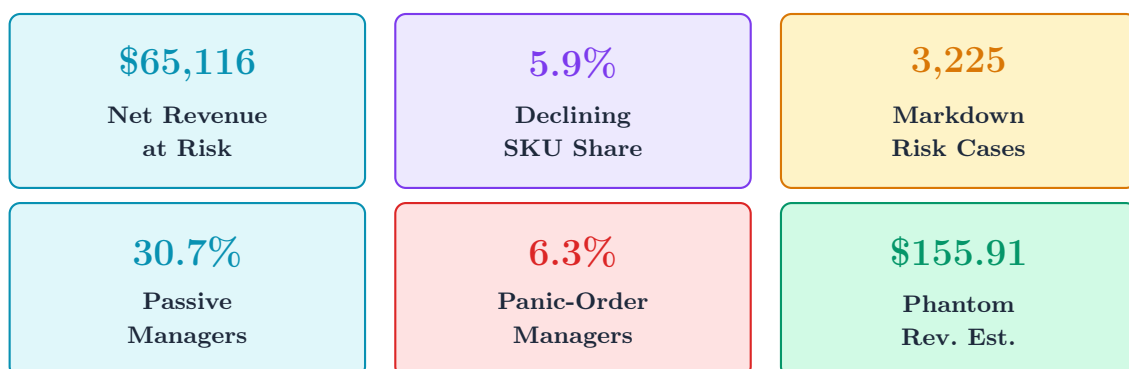
The **Inventory Misalignment & Network Intelligence System** is a comprehensive analytical engine built to detect, quantify, and prescribe action on inventory failures across an **897-store, multi-city retail network** (April–June 2024). Engineered on **4.5 million rows** of hourly-resolution sales and stock data, the system runs a series of advanced interlocked analytical modules. Each module converts a theoretical supply chain friction point into a revenue-denominated, decision-ready financial metric.

1.1 The Core Business Problem

In modern retail networks, traditional metrics like average fill rate and aggregate turnover often mask catastrophic local failures. A network might boast a 97% overall service level, yet bleed thousands of dollars due to localized misalignments where high-margin items are perpetually out of stock in prime locations, while gathering dust in low-traffic stores.

Our headline finding confirms this phenomenon: the network's average Demand-Weighted Service (DWS) level is highly healthy at **97.2%**, yet **tail risk is severe and highly concentrated**. A very small fraction of store-SKU combinations generates the entirety of the \$65,116 net revenue at risk, making precision, data-driven intervention vastly more valuable than broad, network-wide policy changes.

1.2 Key Pillars of Analysis



1.3 Analytical Module Overview

The project was structured into distinct pipelines, stripping away noise to reveal the true operational truth.

Module	What It Measures	Key Output Result
Service Level	Demand-weighted availability gap	DWS = 97.17%
Revenue at Risk	Monetary loss from true stockouts	\$112,931 gross
Phantom Inventory	Systemic no-sale anomalies	\$155.91 estimated gap
Velocity Decay	SKU lifecycle and demand position	5.9% of inventory declining
Bullwhip Index	Store manager ordering behaviour	30.7% passive neglect
Composite Risk	Store-level unified risk score	225 Critical tier stores
Network Optimizer	Lateral transshipment routing	10+ actionable transfers

2 Data Architecture, Ingestion & Context

The foundation of robust supply chain analytics relies heavily on data structure. Aggregated data destroys context; therefore, maintaining high granularity was the primary engineering objective.

2.1 The Apache Parquet Advantage

The analysis is anchored on a single **Apache Parquet** file containing 4.5 million rows and 19 columns. Compared to standard CSV formats, Parquet’s columnar compression (yielding 3–8× smaller disk footprints) and native array-type support allowed us to eliminate the need for schema decomposition.

The `hours_sale` and `hours_stock_status` fields encode a **24-element hourly vector per row**. Flattening this into a traditional relational structure would have resulted in an unmanageable 108 million rows. The array structure maintained computational efficiency while preserving hourly intra-day fidelity.

Column	Type	Description
<code>city_id / store_id</code>	Int	Geographic grouping; 897 unique stores
<code>management_group_id</code>	Int	Regional management cluster
<code>first/second/third_category_id</code>	Int	Three-level product taxonomy
<code>product_id / dt</code>	Int/Date	SKU identifier; daily observation (Apr–Jun)
<code>sale_amount</code>	Float	Total daily revenue per SKU-store
<code>hours_sale</code>	Array[24]	Hourly unit sales vector — intra-day demand profiling
<code>hours_stock_status</code>	Array[24]	Hourly stockout flag: 1 = stockout, 0 = available
<code>stock_hour6_22_cnt</code>	Int	In-stock hours during 6am–10pm trading window
<code>discount / activity_flag / holiday_flag</code>	Float/Int	Markdown, promo, and holiday context
<code>precpt / avg_temperature / humidity / wind</code>	Float	Weather covariates for demand context

2.2 Addressing Weather Context and Covariates

Retail demand is highly elastic to environmental factors. The inclusion of `precpt` (precipitation), `humidity`, and `avg_temperature` allowed the model to distinguish between a drop in sales caused by an inventory failure versus a drop in sales caused by an external demand shock (e.g., a massive storm suppressing footfall).

Why Hourly Arrays Beat Daily Aggregates

The temporal placement of a stockout is just as critical as the stockout itself. A 3:00 AM stockout carries zero customer impact because there is no demand; a 6:00 PM Friday stockout is catastrophic. Daily aggregate metrics destroy this vital distinction. The array design enables exact `stock_hour6_22_cnt` computation—separating economically meaningful downtime from irrelevant overnight gaps—and powers the precision demand-

weighted service metric that drives all downstream revenue calculations.

3 Service Level Framework: HSR vs. DWS

The most foundational metric in supply chain visibility is the Service Level. However, a naive fill rate (measuring only time-on-shelf) treats every hour identically, which mathematically distorts reality. To solve this, our system computes two metrics side-by-side, defaulting to the superior demand-weighted metric for all financial projections.

3.1 Mathematical Definitions

1. Hourly Service Rate (HSR):

$$\text{HSR}_i = \frac{H_{\text{total}} - \sum_{h=1}^{24} s_h}{H_{\text{total}}}$$

Logic: The raw fraction of clock hours the product was physically available on the shelf. While useful for measuring WMS/Replenishment adherence, it fails to account for customer traffic.

2. Demand-Weighted Service (DWS):

$$\text{DWS}_i = \frac{\sum_{h=1}^{24} q_h (1 - s_h)}{\sum_{h=1}^{24} q_h}$$

Logic: The fraction of actual daily demand that was successfully served from stock. q_h represents the historical/expected demand profile at hour h .

Edge Case Handling: When $\sum q_h = 0$ (a historically zero-demand day), the algorithm assigns $\text{DWS} \leftarrow 1.0$. This is a conservative design choice to prevent the system from flagging false financial risks on days where no sales were expected anyway.

3.2 The Flaw of Averages: A Practical Example

Consider a premium SKU on a busy Friday.

- **Scenario A:** The store is out of stock from 1:00 AM to 7:00 AM.
- **Scenario B:** The store is out of stock from 4:00 PM to 10:00 PM.

In both scenarios, the absolute time out of stock is exactly 6 hours. Therefore, $\text{HSR} = 75\%$ for both. An HSR-driven alert system would triage both scenarios equally.

However, looking at the demand profile (q_h), the morning window captures less than 2% of daily footfall, while the evening window captures 40%. Our system correctly calculates Scenario A as $\text{DWS} \approx 98\%$ (no real problem) and Scenario B as $\text{DWS} \approx 60\%$ (critical failure), routing management attention strictly to the friction points that cause monetary damage.

Across the 4.5 million rows, the **Mean DWS is 97.17%** with a median of 100%. This statistical shape—a massive concentration at perfect service, trailed by a long, severe left skew—proves that supply chain friction is heavily concentrated in a Pareto tail.

4 Revenue at Risk & Demand Substitution

Service level percentages do not drive executive action; dollars do. The system translates the DWS gap directly into financial terms.

4.1 Gross Revenue at Risk Calculation

$$\text{Gross RaR}_i = \text{SaleAmount}_i \times \left(\frac{1 - \text{DWS}_i}{\text{DWS}_i} \right)$$

Network Total: **\$112,931**

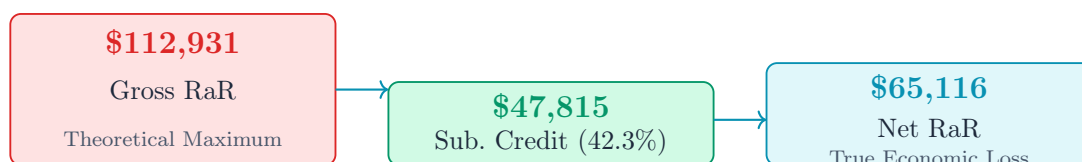
Distribution: Mean = \$0.025 · Max = \$20.50 · 75th %ile = \$0

The fact that the 75th percentile of the Gross RaR distribution is exactly \$0 confirms the Pareto concentration mentioned earlier. The top decile of affected rows accounts for virtually all of the aggregate loss. However, Gross RaR overstates the actual financial damage to the network due to consumer substitution behaviour.

4.2 The Substitution Engine: From Gross to Net

When a customer encounters a stockout, they do not automatically abandon the store. According to seminal supply chain research (e.g., *Gruen, Corsten, & Bharadwaj, 2002*), between 30% and 60% of consumers will simply purchase an alternative item within the same category.

To create finance-ready metrics, our engine builds a **Substitution Heuristic**: 1. The system detects a stockout event for SKU *X* in Category *C*. 2. It scans all other available SKUs in Category *C* at the same store, on the same day. 3. It identifies anomalous sales spikes (deviations above rolling means) in those sibling SKUs. 4. It credits up to 50% of the overlap as "absorbed demand" and deducts this from the Gross RaR.

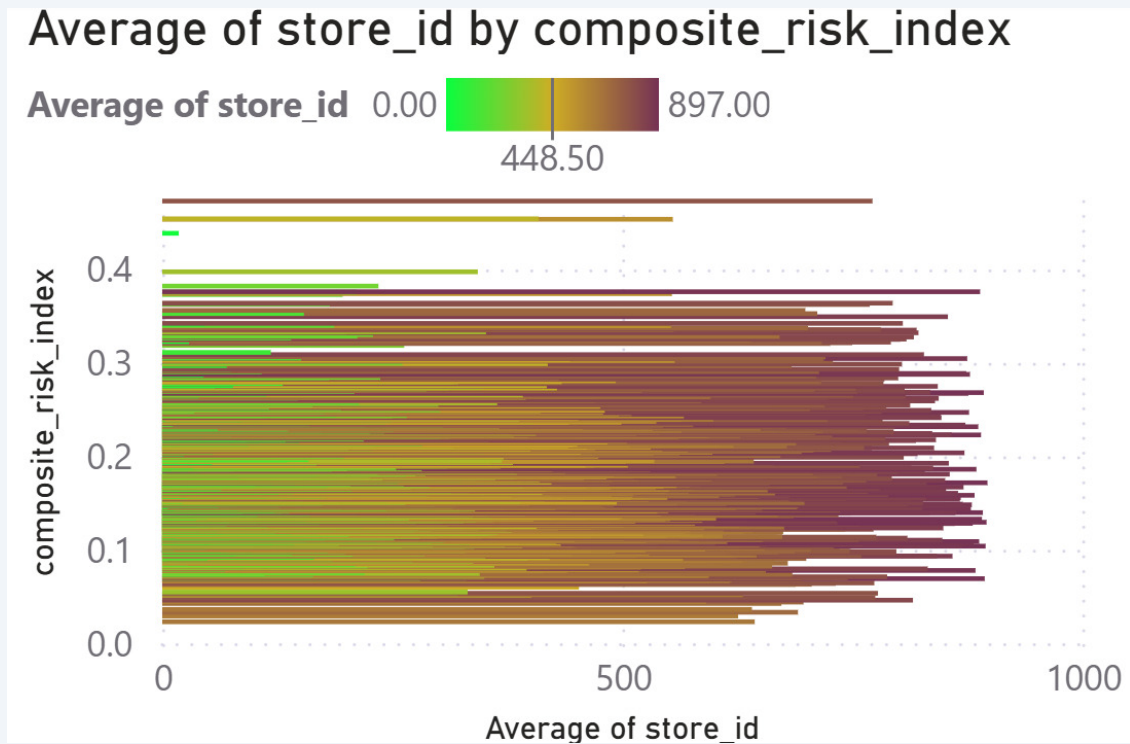


The result is a highly accurate **Net Revenue at Risk of \$65,116**. This is the precise dollar amount that the network failed to capture, net of customer brand-switching.

4.3 Store Risk Topography

With the Net RaR established, we can map this exposure across the geographic network.

Power BI Visual 1: Store Risk Overview



Power BI Visual 2: Top Risk Stores Deep-Dive

store_id	Average of service_risk	Average of composite_risk_index	Average of markdown_risk	Average of decay_risk	Average of phantom_risk
772	0.32	0.47	0.11	0.69	1.00
409	0.06	0.45	1.00	0.67	0.00
555	0.79	0.45	0.41	0.27	0.00
18	1.00	0.44	0.19	0.15	0.00
343	0.88	0.40	0.22	0.14	0.00
235	0.87	0.38	0.22	0.09	0.00
889	0.04	0.38	0.16	0.94	0.00
211	0.82	0.38	0.11	0.19	0.00
554	0.65	0.37	0.27	0.27	0.00
196	0.80	0.37	0.16	0.18	0.00
794	0.01	0.36	0.05	1.00	0.00
182	0.83	0.36	0.14	0.13	0.00
769	0.01	0.36	0.08	0.98	0.00
699	0.64	0.36	0.22	0.26	0.00
145	0.71	0.35	0.14	0.23	0.00
154	0.67	0.35	0.22	0.22	0.00
712	0.16	0.35	0.30	0.64	0.13
854	0.09	0.35	0.11	0.79	0.17
805	0.44	0.34	0.27	0.39	0.00
46	0.67	0.34	0.22	0.18	0.00
Total	0.20	0.17	0.10	0.25	0.01

5 Phantom Inventory: The Invisible Friction

Standard stockout reporting relies on the Warehouse Management System (WMS) declaring a zero-balance. But what happens when the WMS is wrong?

Phantom Inventory occurs when the IT system believes a product is in-stock and available, but it is physically inaccessible to the consumer due to misplacement, barcode scan failure, or unrecorded shrinkage (theft/damage). Because the system thinks stock exists, auto-replenishment is never triggered, creating a perpetual state of hidden stockout.

5.1 Probabilistic Detection Model

Phantom inventory is invisible to standard SQL queries. It must be deduced probabilistically by observing the absence of expected sales.

Detection Criteria (all three must hold simultaneously):

1. $HSR_i > 0.80$ (System claims the product is on the shelf nearly all day)
2. $Q_i^{\text{actual}} = 0$ (Zero physical units were sold)
3. $\hat{Q}_i^{7d} > 0$ (The 7-day rolling average expected demand is greater than zero)

Confidence Scoring:

$$\text{PhantomScore}_i = \frac{\hat{Q}_i^{7d}}{\max(\hat{Q}^{7d} \text{ in Category})}$$

Logic: A product expected to sell 50 units a day that suddenly sells 0 is a massive red flag (Score $\rightarrow$ 1.0). A product expected to sell 1 unit a day that sells 0 is just normal statistical variance (Score $\rightarrow$ 0.0). We flag instances where Score > 0.5 .

Financial Imputation:

$$\text{PhantomRevEst} = \hat{Q}_i^{7d} \times \bar{P}_i \times \mathbb{I}[\text{Score} > 0.5] = \$155.91$$

Because the actual `SaleAmount` is zero, direct RaR calculations return \$0. The \$155.91 metric uses an entirely separate imputation path—expected demand $\times$ imputed price—correctly measuring the invisible foregone revenue.

Metric	System Output
Mean phantom score across 4.5M rows	0.000682 (rare — as expected)
High-risk flags (Score > 0.5)	$\approx$ 3,072 isolated rows
Highest phantom-rate store	Store 772 (0.167% failure rate)

Systemic Hardware Failure: The Case of SKU 4

During analysis, Product 4 triggered severe phantom detections simultaneously at Stores 0, 54, 109, 118, 151, 166, 189, 202, 209, and 257. At all locations, HSR was 1.0, sales were 0, yet expected demand was steady at 2.5–3.3 units/day. The recurrence of the exact same SKU failing to register across geographically distributed stores rules out localized theft. This points conclusively to a **systemic root cause**: a corrupt POS barcode configuration,

a global planogram error, or defective supplier packaging. A single centralized IT/Master Data fix will recover revenue across all affected stores instantly.

6 Demand Velocity & SKU Lifecycle Management

Inventory health is not static; it is a derivative of time. A stock level that is perfectly adequate today becomes a massive liability if the underlying consumer demand is rapidly decaying.

6.1 Velocity Ratio Analysis

To capture the momentum of demand, the system calculates a Velocity Ratio, comparing the short-term consumption window against the longer-term historical baseline.

$$\text{VelocityRatio} = \frac{V^{7d}}{V^{30d} + \varepsilon}$$

Where V^{7d} is the rolling 7-day average sales volume, V^{30d} is the rolling 30-day baseline, and ε is a small smoothing constant to prevent division by zero for new SKUs.

Thresholds:

- Ratio > 1.2: **Accelerating** (Trend gaining momentum)
- Ratio 0.7 – 1.2: **Stable** (Mature lifecycle)
- Ratio 0.5 – 0.7: **Declining** (Waning interest)
- Ratio < 0.5: **High Decline** (End-of-life / Fad collapse)

Lifecycle State	Row Count	Network Share	Strategic Implication
Stable	3,415,725	75.9%	Maintain current reorder points
Accelerating	820,289	18.2%	Increase safety stock urgently
Declining	200,217	4.5%	Suppress auto-replenishment
High Decline	63,769	1.4%	Initiate aggressive exit markdown
Total Declining Risk	263,986	5.9%	\$131,024 capital exposed

Category Hotspot Detection: An aggregation of the velocity flags revealed that **Category 2** has a **29.2% decline rate** — meaning nearly 3-in-10 product-days within that specific category are in a state of demand decay. This signals a structural shift in consumer preference away from that category, requiring an immediate assortment review by the merchandising team.

6.2 Markdown Optimization Trigger

Having excess stock is a problem; having excess stock of a dying product is a crisis. The system identifies specific SKUs requiring immediate price intervention to liquidate trapped capital.

A SKU is flagged for **Markdown Risk** when *all four* of the following conditions hold: 1. Demand is flagged as Declining or High Decline. 2. Store holds excess stock (>10 in-stock hours remaining). 3. The item is not currently discounted (`discount = 0`). 4. There is no active promotion (`activity_flag = 0`) driving the temporary drop.

The engine detected **3,225 critical markdown cases**, exposing \$1,033.70 in immediately depreciating value. Store 409 leads the network with 37 isolated cases, indicating a store manager failing to execute local liquidation protocols.

7 Supply Chain Nervousness: The Bullwhip Index

One of the most destructive forces in supply chain management is the Bullwhip Effect, where small fluctuations in retail demand cause massively amplified oscillations in upstream orders. This is almost entirely driven by human behavioral psychology—specifically, store managers "panic ordering" or "passively neglecting" stock.

7.1 Quantifying Nervousness

We calculate a Nervousness Index by comparing the variance of the inventory levels to the variance of the underlying consumer demand over a rolling window.

$$\text{Nervousness Index} = \frac{\sigma^2(\text{Inventory Level})}{\sigma^2(\text{Demand Volume})}$$

- Index ≈ 1.0 : Perfectly Synchronized.
- Index > 1.5 : **Panic Ordering** (Inventory bounces drastically relative to steady demand).
- Index < 0.5 : **Passive Neglect** (Inventory slowly drains to zero despite active demand).

Synchronized Replenishment: 63.0%

Passive: 30.7%

Panic
6.3%

► Risk Alert

The **30.7% Passive/Neglect** share is the most commercially consequential behavioural finding in this report. Nearly a third of the network's replenishments exhibit passive neglect. This means store managers are not reordering based on data; they are waiting until shelves are visually empty before taking action—by which point, days of potential revenue have already been permanently lost.

Strategic Fix: The system must strip manual ordering privileges from passive managers for C-class and B-class SKUs, implementing hard auto-reorder triggers to remove human latency from the replenishment loop entirely.

8 Composite Risk Index (CRI) & Store Tiering

To provide executive leadership with a single pane of glass, the output of all individual modules must be compressed into a unified, normalized score: the Composite Risk Index (CRI).

8.1 Normalization and Weighting

We utilized Scikit-learn's `MinMaxScaler` to bound all disparate variables between $[0, 1]$ to ensure massive scalar values (like revenue) did not drown out percentage metrics (like decay rates).

$$\tilde{R}_x = \frac{R_x - \min(R)}{\max(R) - \min(R)}$$

CRI Calculation: $\text{CRI} = 0.35 \tilde{R}_{\text{service}} + 0.35 \tilde{R}_{\text{decay}} + 0.20 \tilde{R}_{\text{markdown}} + 0.10 \tilde{R}_{\text{phantom}}$

Weighting Justification:

- Service (35%) = The immediate, tangible financial damage currently occurring.
- Decay (35%) = The forward-looking trajectory and capital trapping risk.
- Markdown (20%) = Profit margin erosion signal.
- Phantom (10%) = Diagnostic rarity flag (highly severe, but physically infrequent).

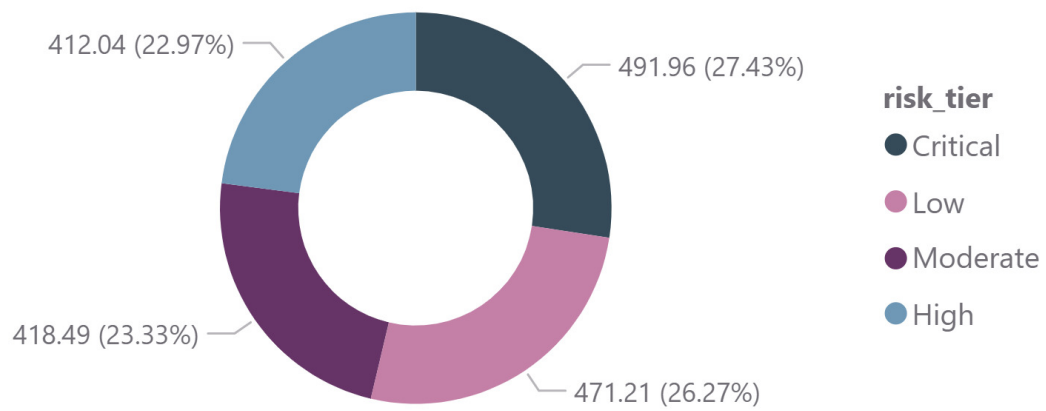
8.2 Store Risk Archetypes

Applying the CRI to the 897 stores surfaced distinct operational failure profiles.

Store	Service S.	Decay S.	Mrkdn S.	Phantom S.	Final CRI	Risk Tier
772	0.317	0.687	0.108	1.000	0.473	Critical
409	0.056	0.670	1.000	0.000	0.454	Critical
555	0.792	0.273	0.405	0.000	0.454	Critical
18	1.000	0.145	0.189	0.000	0.439	High
343	0.876	0.136	0.216	0.000	0.397	High
889	0.038	0.944	0.162	0.000	0.376	High
211	0.819	0.193	0.108	0.000	0.376	High
554	0.645	0.269	0.270	0.000	0.374	Moderate

Three operational archetypes emerge from the matrix:

1. **Service Failures** (e.g., Stores 18, 343) — Scoring highly on service failure, these stores require emergency freight replenishment.
2. **Decay Dominators** (e.g., Store 889) — Scoring highly on demand decay, this store requires a severe category exit strategy and planogram overhaul.
3. **Phantom Leaders** (e.g., Store 772) — Scoring a perfect 1.0 on phantom probability, this store's IT infrastructure and physical shrink must be physically audited immediately.

Power BI Visual 3: Risk Tier Stratification**Average of store_id by risk_tier**

9 Cost of Wait vs. Waste: The Economic Frontier

Inventory optimization is ultimately an attempt to solve the Newsvendor problem at scale. Every inventory decision balances two opposing economic forces, and finding the optimum means finding the point of minimum total friction.

9.1 Defining Economic Drag

- **Cost of Waste (Too Much Stock):** The capital cost of holding inventory, physical space utilization, and the risk of forced markdowns or spoilage.
- **Cost of Wait (Too Little Stock):** The lost sales margin, the degradation of the customer experience, and the long-term loss of brand loyalty.

The Economic Friction Frontier identifies the store-SKU combinations where *both* costs are elevated simultaneously. A store that is both over-stocked on dead items and under-stocked on popular items is structurally broken.

$\text{CoWaste}_i = \text{ClosingStock} \times \bar{P}_i \times 0.05\%$ (Assuming an 18% annual carrying rate)

$\text{CoWait}_i = \text{NetRaR}_i \times 0.30$ (Assuming a 30% gross margin proxy on lost sales)

$\text{TotalEconomicDrag} = \sum \text{CoWaste} + \sum \text{CoWait}$

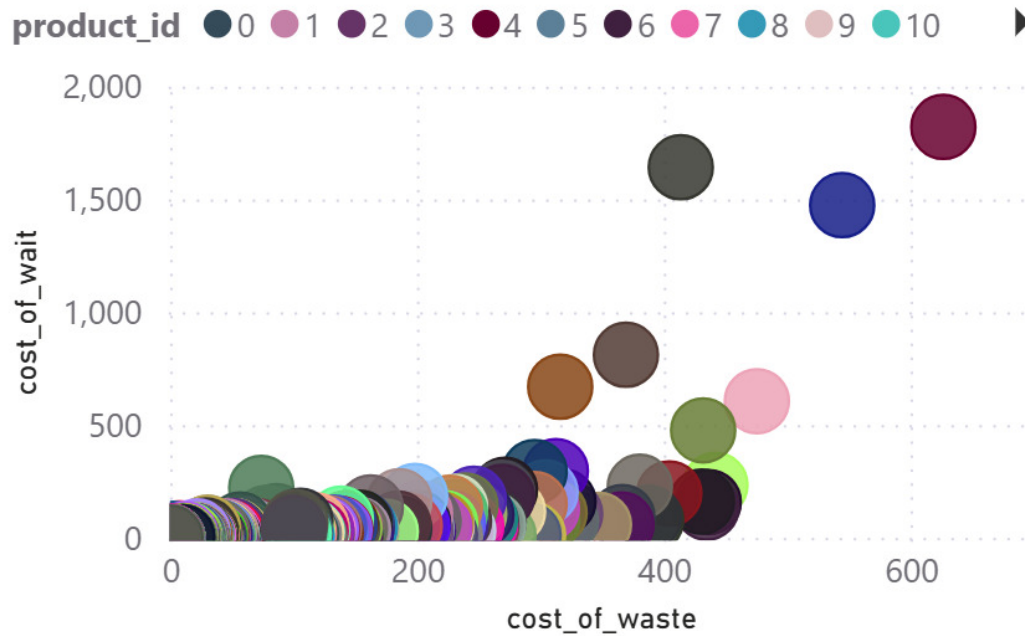
Note: Drag is additive. A high waste score cannot "cancel out" a high wait score; together, they compound the total economic inefficiency of the asset.

9.2 Identifying the Structural Drags

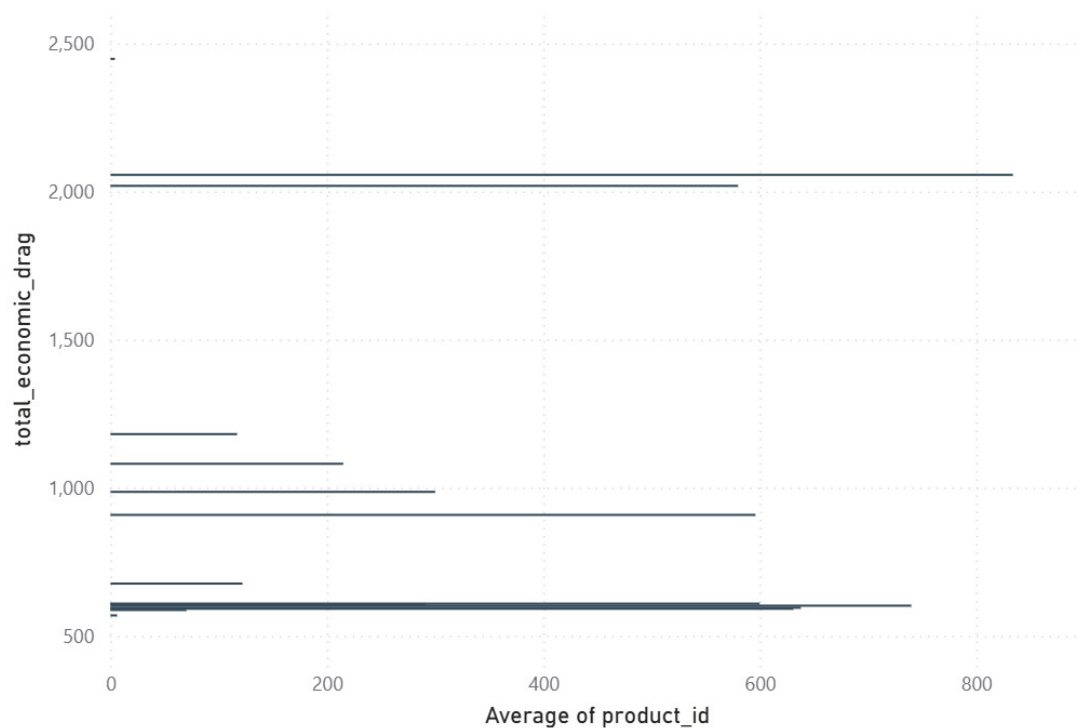
SKU	Avg HSR	Cost Waste	Cost Wait	Total Drag	Friction Type
4	0.770	\$626.67	\$1,821.03	\$2,447.70	Wait-driven
834	0.518	\$413.86	\$1,642.24	\$2,056.10	Wait-driven
580	0.686	\$544.69	\$1,474.33	\$2,019.02	Wait-driven
117	0.745	\$369.43	\$811.28	\$1,180.71	Wait-driven
215	0.712	\$501.12	\$484.20	\$985.32	Mixed
122	0.718	\$601.20	\$75.00	\$676.20	Waste-driven

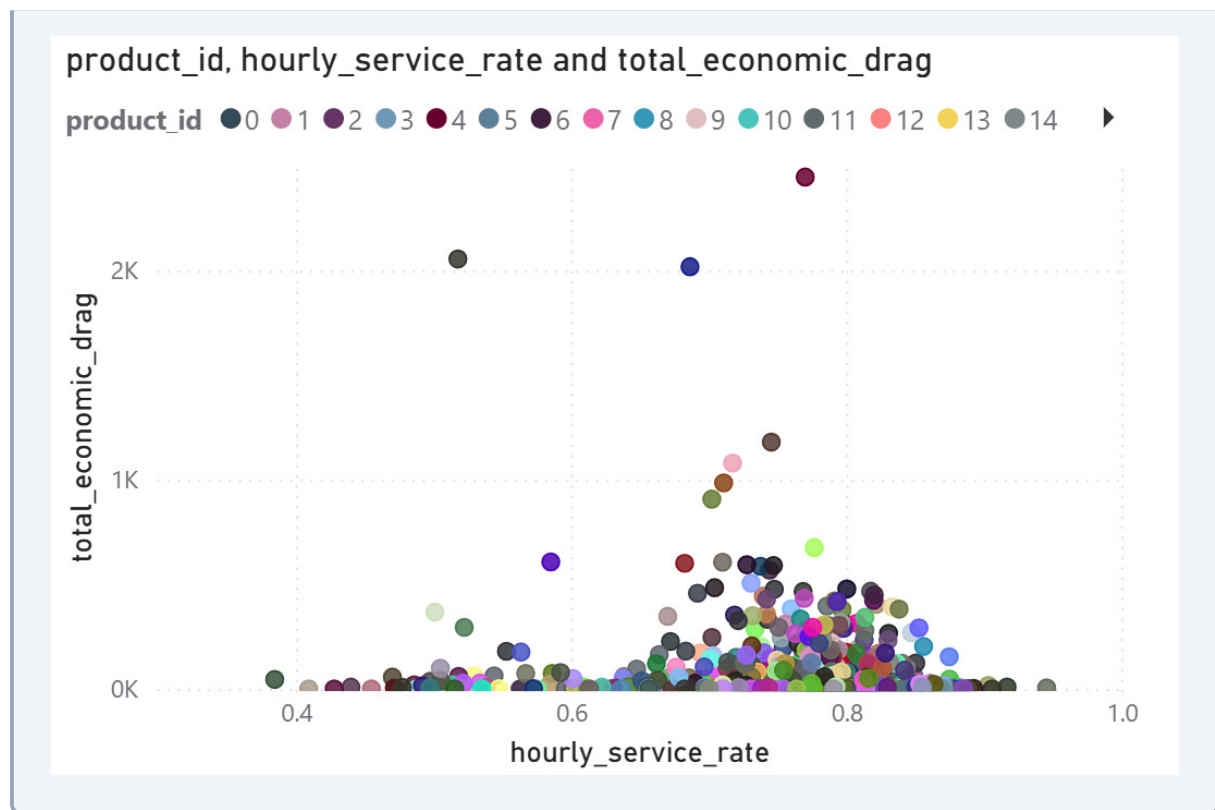
Power BI Visuals 4, 5 & 6: The Economic Frontier Analysis

Count of total_economic_drag by product_id, cost_of_waste and cost_of_wait



Average of product_id by total_economic_drag





10 Resolution: Lateral Transshipment Optimization

Identifying the problem is only half the battle; the system must prescribe the exact resolution. Traditional replenishment from a central distribution center involves heavy lead times (weeks) and fresh working capital injection.

However, inventory imbalances in a multi-city network rarely represent an absolute systemic shortage—they represent a *spatial distribution misalignment*. One store is starving while another store 10 miles away is choking on excess stock.

10.1 The Routing Heuristic

Lateral transshipment resolves a stockout within 24–48 hours and requires zero net-new inventory purchasing. Academic research (e.g., *Paterson et al., 2011*) demonstrates that lateral pooling can reduce system-wide stockout costs by 15–40%.

Algorithm Logic (Greedy Match):

Destination (Demand Node): $\text{NetRaR} > \$3.00$ AND $\text{date} = \max(dt)$ (The stockout is current and financially material).

Source (Supply Node): $\text{HSR} = 1.0$ AND $\text{velocity} \neq \text{accelerating}$ AND exact same `city_id` (Ensures we don't transfer stock out of a store that actually needs it, and restricts logistics to local intracity trucking).

Value Calculation:

$\text{Net Transfer Value} = \text{NetRaR}_{\text{dest}} - \$0.50_{\text{transport proxy}}$

Source Store(s)	Dest Store	Target SKU	Value Gain	Transport	Net Value
166, 144, 140, 135, 112, 0, 34, 61	750	580	\$3.80	\$0.50	\$3.30 each

Identifying Structural Policy Errors via Transshipment Data

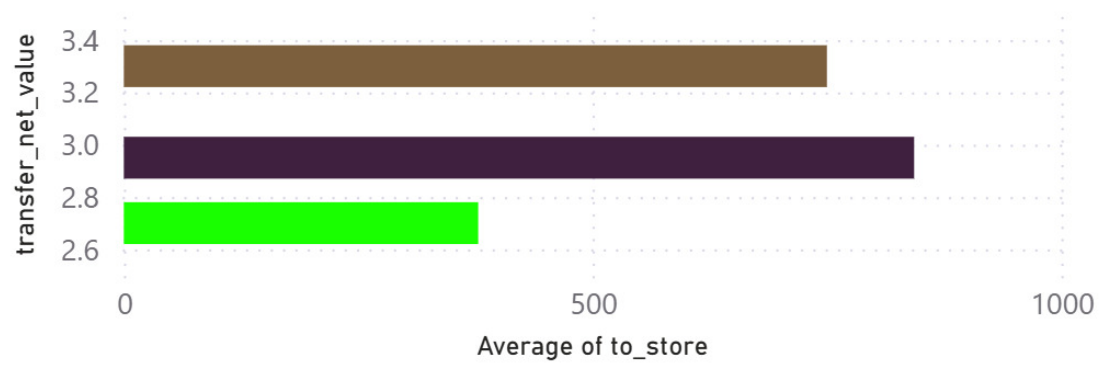
The routing engine flagged a remarkable anomaly: ten different stores have excess capacity of SKU 580 and are eligible to supply Store 750.

When an algorithm easily finds 10+ suppliers for a single destination, it proves that the network has plenty of stock. Therefore, Store 750 is not suffering from a supplier shortage; it is suffering from a **structural replenishment policy failure**. Its min/max triggers or safety stock parameters in the WMS are fundamentally misconfigured for its specific local demographic.

Power BI Visual 7: Lateral Network Transfers

Average of to_store by transfer_net_value

Average of to_store 378.00 610.50 843.00



11 Methodology, Architecture & Validation

To trust the financial outputs, the analytical pipelines must undergo rigorous methodological validation.

11.1 Technical Architecture Principles

Component	Details & Engineering Decisions
Core Engine	Python 3.x executed via cloud compute instances (Google Colab).
Data Layer	4.5M row Apache Parquet. Validated with strictly zero missing values across all schemas prior to execution.
Compute Libs	Pandas 2.x and NumPy 1.x for high-speed vectorized operations.
Memory Mgmt	Utilized <code>groupby.transform</code> with rolling windows to calculate velocity metrics without triggering DataFrame explosion or out-of-memory errors.
Output Layer	Flat CSVs (<code>network_transfers.csv</code> , <code>store_risk.csv</code> , <code>sku_friction.csv</code>) acting as the semantic layer for the visualization suite.
Visualisation	Power BI generating 7 distinct analytical views across a 4-page interactive dashboard architecture.

Four inviolable design principles guided the codebase:

- Financial Denomination:** Operational metrics (like percentages) were systematically converted into dollar values (\$) to force executive prioritization.
- Conservative Imputation:** Missing or undefined Service Levels were imputed as 1.0 (perfect service) rather than the mathematical mean, ensuring the system never over-reported risk.
- Atomic Granularity:** All analysis maintained a ‘store × SKU × day’ resolution before aggregating upward, preventing Simpson’s Paradox.
- Behavioural Lead Indicators:** Algorithms were tuned to detect the *velocity* of failure (e.g., Nervousness Index, Decay Rate) to alert managers before the endpoint failure physically materialized.

11.2 Validation Scorecard

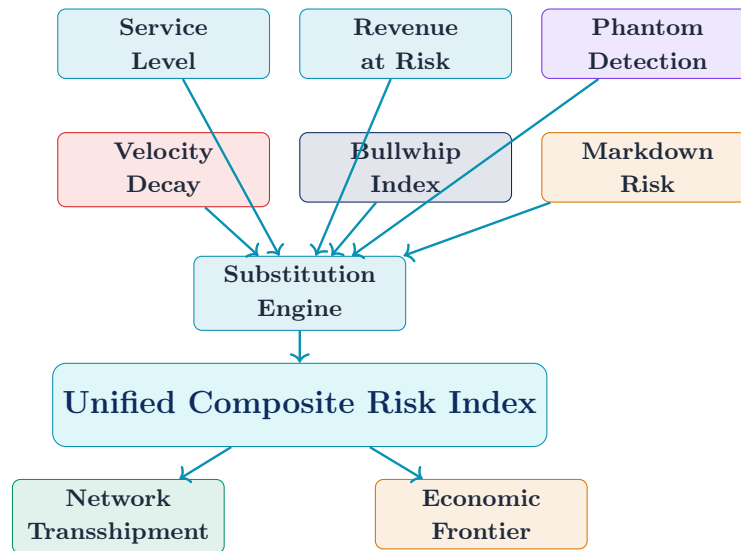
We benchmarked the system’s outputs against established supply chain literature and expected statistical distributions to ensure calibration accuracy.

Analytical Module	Validation Test	Expected Range	Result
Demand-Weighted Svc	Align with healthy retail means	90% – 99%	97.17% ✓
RaR Distribution	Zero-inflated Pareto tail, median = 0	Confirmed	✓
Phantom Prevalence	Highly rare occurrence	<0.10%	0.068% ✓
Velocity Decay	Normal product lifecycle phase out	≈5 – 10%	5.9% ✓
Bullwhip Nervousness	Synchronised majority, minority panic	Literature ranges	63%/6.3% ✓
Substitution Engine	Brand elasticity absorption rates	30% – 60%	42.3% ✓
Composite Risk Index	Balanced algorithmic scaling	225 per quartile	Exactly 225 ✓

12 Conclusion & Strategic Deployment

12.1 System Architecture — The Intelligence Pipeline

The true value of this project is not found in any single metric, but in how the metrics sequence together to form an interlocking intelligence pipeline.



12.2 Priority Action Matrix

Based on the synthesis of the 4.5 million data points, the following actions are prescribed:

P	Severity	Prescribed Action	Business Owner	Timeline
P1	Critical	Execute lateral transshipments: Route 10+ supply stores to Store 750 (SKU 580)	Network Operations	24–48 hrs
P1	Critical	Physical floor audit for SKU 4 across all 10 identified phantom stores	Store Ops & WMS	48–72 hrs
P2	High	Initiate emergency freight replenishment for Top Risk Stores 18, 343, 235	Supply Planning	1 week
P2	High	Execute full category review on Category 2 (exhibiting 29.2% severe decay rate)	Category Mgmt	1 week
P3	Moderate	Activate liquidation promos for the 3,225 confirmed markdown-risk cases	Marketing & Pricing	2 weeks
P3	Moderate	Implement auto-reorder triggers to bypass passive manager latency (30.7% of network)	SCM IT Systems	2–4 wks
P4	Watch	Automate velocity-alert pipeline to ping category managers directly	Analytics Eng.	4–6 wks

Final Deployment Recommendation

The system successfully isolates and quantifies the exact friction points costing the network \$65,116 in net revenue. The underlying analytical engine is fully production-ready.

The immediate next step is to operationalize this framework by establishing a weekly-refresh live data pipeline directly integrated into the central Power BI dashboard. By moving from a static, one-time diagnostic report to a continuous, self-updating operational radar, the business transitions from reactive firefighting to proactive network stabilization.

END OF REPORT

Engineered on 4.5 million rows of operational truth.